



MOVES: INTERVIEW

What technologies are available for thermal management, and how do they compare?

Thermal management technologies include liquid immersion cooling, which uses dielectric coolant to stabilise cell temperatures, and thermal foam for effective heat dissipation. However, many manufacturers still rely on potting materials without exploring broader options. Tailoring the thermal solution to a pack's specific needs is essential for optimal safety and performance.

Why are thermal materials particularly important in Indian two-wheelers and three-wheelers?

Given India's reliance on air cooling for two- and three-wheelers, selecting appropriate thermal materials is vital for safety. Detailed engineering analysis of cell type, heat generation, and usage profiles ensures that temperature rise (ΔT) remains within 4°C , a standard upheld in our battery packs.

What steps are necessary to ensure the correct selection of materials?

To ensure optimal performance, a detailed cell profile under various charge and discharge conditions is essential. When integrating it into the battery pack, a comprehensive CFD (computational fluid dynamics) analysis must be conducted to evaluate heat generation under different scenarios. By analysing extreme charge and discharge conditions, the required heat dissipation can be determined to maintain safety and efficiency.

What type of battery management system ensures effective cell balancing while adhering to safety protocols?

Choosing the right battery management system (BMS) involves ensuring safety protocols align with operational needs. For high discharge rates (above 100 to 120 amps), contactor-based mechanisms are suitable, while MOSFET-based safety suffices for lower rates. AISIL compliance is critical for automotive-grade BMS, yet cost constraints limit adoption among tier-two manufacturers. This disparity highlights the prioritisation of safety by tier-one OEMs versus the cost-driven compromises of others. Globally, generation 3 BMS systems—standard in Europe—outperform domestic options with superior safety features, enhanced balancing currents, and precise state of charge measurement, underscoring the gap in standards. **EFY**



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